

1. Open your Windows or Linux system, use the command line to assign a static IPv4 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.7417]
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C:\WINDOWS\system32>netsh interface ipv4 show interfaces

Idx  Met      MTU      State      Name
-----
1    75    4294967295  connected  Loopback Pseudo-Interface 1
6    55      1500    connected  WiFi
4    5      1500    disconnected Ethernet
16   25      1500    disconnected Local Area Connection* 1
12   65      1500    disconnected Bluetooth Network Connection
9    25      1500    disconnected Local Area Connection* 2
7    35      1500    connected  VMware Network Adapter VMnet1
20   35      1500    connected  VMware Network Adapter VMnet8

C:\WINDOWS\system32>netsh interface ip set address name="WiFi" static 192.168.1.12 255.255.255.0 192.168.1.1

C:\WINDOWS\system32>ipconfig
```

```
Wireless LAN adapter WiFi:

Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2001:db8:1::100
IPv6 Address. . . . . : 2402:a00:408:8f67:7ec3:d8aa:6308:d824
IPv6 Address. . . . . : fd00:1:1::100
Temporary IPv6 Address. . . . . : 2402:a00:408:8f67:a832:19b3:24cd:3972
Link-local IPv6 Address . . . . . : fe80::bcff:c2db:3220:39b7%6
IPv4 Address. . . . . : 192.168.1.12
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1%6
                             192.168.1.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\WINDOWS\system32>
```

2. **Configure a local IPv4 address on your system and confirm connectivity by pinging another device on your LAN that also has a local IPv4 address.**

```
Wireless LAN adapter WiFi:
Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2001:db8:1::100
IPv6 Address. . . . . : 2402:a00:400:8f67:7ec3:d8aa:6308:d824
IPv6 Address. . . . . : fd00:1:1::100
Temporary IPv6 Address. . . . . : 2402:a00:400:8f67:a832:19b3:24cd:3972
Link-local IPv6 Address . . . . . : fe80::bcff:c2db:3220:39b7%6
IPv4 Address. . . . . : 192.168.1.10
Subnet Mask . . . . . : 255.255.0
Default Gateway . . . . . : fe80::1%6
192.168.1.1

Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\WINDOWS\system32>ping -4 192.168.1.71

Pinging 192.168.1.71 with 32 bytes of data:
Reply from 192.168.1.10: Destination host unreachable.
Reply from 192.168.1.10: Destination host unreachable.
Reply from 192.168.1.10: Destination host unreachable.
Reply from 192.168.1.10: Destination host unreachable.

Ping statistics for 192.168.1.71:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\WINDOWS\system32>
```

3. **Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Document the steps and the observed address behaviour in a text file.**

- Steps which I performed to check assigned IP is changed or not after rebooting. Here, as follows:
- Opened Command Prompt.
- Assigned a temporary IPv4 address:
192.168.1.100
- Verified the address using ipconfig
- Restarted the computer after disconnecting from the internet.
- Checked the IPv4 address again.
- And I found that before restarting IPv4 address was : 192.168.1.100
- And after restarting IPv4 address remains same which is 192.168.1.100

4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv4 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv4 address.

- Here is the python script which has restaurant names and it's IPv4 addresses:

```
#Zomato-style Restaurant Search using IPv4 Addresses
```

```
Restaurants = {  
    {"name": "Dominos Pizza", "ip": "192.168.1.10"},  
    {"name": "McDonald's", "ip": "192.168.1.11"},  
    {"name": "Burger King", "ip": "192.168.1.12"},  
    {"name": "KFC", "ip": "192.168.1.13"},  
    {"name": "Subway", "ip": "192.168.1.14"},  
    {"name": "Pizza Hut", "ip": "192.168.1.15"},  
    {"name": "Barbeque Nation", "ip": "192.168.1.16"},  
    {"name": "Haldiram's", "ip": "192.168.1.17"}  
}  
  
#Get user input  
Search = input("Enter part of the restaurant name: ").lower()  
  
#Search for matches  
For restaurant in restaurants:  
    If search in restaurant ["name"].lower():  
        Matches.append(restaurant)  
  
#Display results  
If matches:  
    print("\Matching Restaurants:")  
    For restaurant in matches:  
        print(f'{restaurant["name"]} --> IPV4 address : {restaurant["ip"]}')  
Else:  
    print("No matching restaurant found.")
```

Example output:

Enter part of the restaurant name : pizza

Matching Restaurants:

Dominos Pizza -- > IPV4 address: 192.168.1.10

Pizza Hut -- > IPV4 address: 192.168.1.15

5. Use ChatGPT or Copilot to generate a sample IPv4 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv4 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

Title: IPv4 Subnetting Scenario and Manual IP Configuration

Part 1: AI-Generated IPv4 Subnetting Scenario

Network Address: 192.168.10.0/24

The network is divided into three subnets:

Subnet 1:

- Department: Administration
- Network Address: 192.168.10.0/26
- Subnet Mask: 255.255.255.192
- Usable Host Range: 192.168.10.1 – 192.168.10.62
- Broadcast Address: 192.168.10.63

Subnet 2:

- Department: Sales
- Network Address: 192.168.10.64/26
- Subnet Mask: 255.255.255.192
- Usable Host Range: 192.168.10.65 – 192.168.10.126
- Broadcast Address: 192.168.10.127

Subnet 3:

- Department: IT
- Network Address: 192.168.10.128/26
- Subnet Mask: 255.255.255.192
- Usable Host Range: 192.168.10.129 – 192.168.10.190
- Broadcast Address: 192.168.10.191

Part 2: Manual IPv4 Configuration

Selected Subnet:

IT Department

Network: 192.168.10.128/26

Assigned IPv4 Address:

192.168.10.130

Subnet Mask:

255.255.255.192

Default Gateway:

192.168.10.129

Preferred DNS Server:

8.8.8.8

Alternate DNS Server:

8.8.4.4

Configuration Steps

1. Open Control Panel.
2. Go to Network and Internet > Network and Sharing Center.
3. Click Change adapter settings.
4. Right-click the active Ethernet or Wi-Fi adapter and select Properties.
5. Select Internet Protocol Version 4 (TCP/IPv4) and click Properties.
6. Choose "Use the following IP address."
7. Enter:
 - IP Address: 192.168.10.130
 - Subnet Mask: 255.255.255.192
 - Default Gateway: 192.168.10.129
8. Enter the DNS servers:
 - Preferred DNS: 8.8.8.8

- Alternate DNS: 8.8.4.4

9. Click OK and close all windows.

10. Open Command Prompt and verify the configuration by running: `ipconfig`